How Does CNN Depth Affect Feature Learning? A Comparative Study of Shallow, Medium and Deep Networks on CIFAR-10

https://github.com/MaheshGollapelli-24165281/cnn-depth-analysis

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ABSTRACT

Three CNN models (shallow, medium, and deep) were evaluated on the CIFAR-10 dataset to analyse how network depth influences feature learning and classification performance.
The results show that deeper networks learn more complex and abstract features, improving accuracy, although the benefits diminish at higher depths and introduce trade-offs such as increased computational cost and risk of overfitting.

INTRODUCTION

Machine learning plays a critical role in analysing complex data across domains such as healthcare, finance, and computer vision. Among various machine learning models, Convolutional Neural Networks (CNNs) are particularly effective for image-based tasks.

CNNs work by automatically learning hierarchical representations of images through multiple layers. One important design factor is the depth of the network. A shallow network captures basic features, while deeper networks can learn more abstract and complex patterns.

This tutorial aims to answer the question: how does CNN depth affect feature learning and classification performance? By comparing shallow, medium, and deep convolutional neural networks on the CIFAR-10 dataset, this study investigates how increasing network depth influences the complexity of learned features and overall model accuracy.

BACKGROUND THEORY

Convolutional Neural Networks consist of layers that apply filters to input images to detect patterns. Early layers typically capture simple features such as edges and textures, while deeper layers combine these features into more complex representations.

Network depth refers to the number of layers in the model. Increasing depth allows the network to learn hierarchical feature representations, where complex patterns are built progressively from simpler ones.

However, deeper networks are not always better. They require more computational resources and may overfit if not properly regularised. While increasing CNN depth improves representational power, deeper networks introduce optimisation challenges such as the vanishing gradient problem, where gradients become too small to effectively update earlier layers. This can slow or prevent learning in very deep architectures.

Modern approaches such as Residual Networks (ResNets) address this issue using skip connections, allowing gradients to flow more effectively through the network. Additionally, deeper models increase the risk of overfitting, particularly on relatively small datasets such as CIFAR-10, making techniques such as dropout and batch normalisation essential.

DATASET AND EXPERIMENTAL SETUP

The CIFAR-10 dataset was used for this experiment. It contains 60,000 colour images across 10 classes, including animals and vehicles. Each image has a resolution of 32×32 pixels.

Three CNN models were implemented:

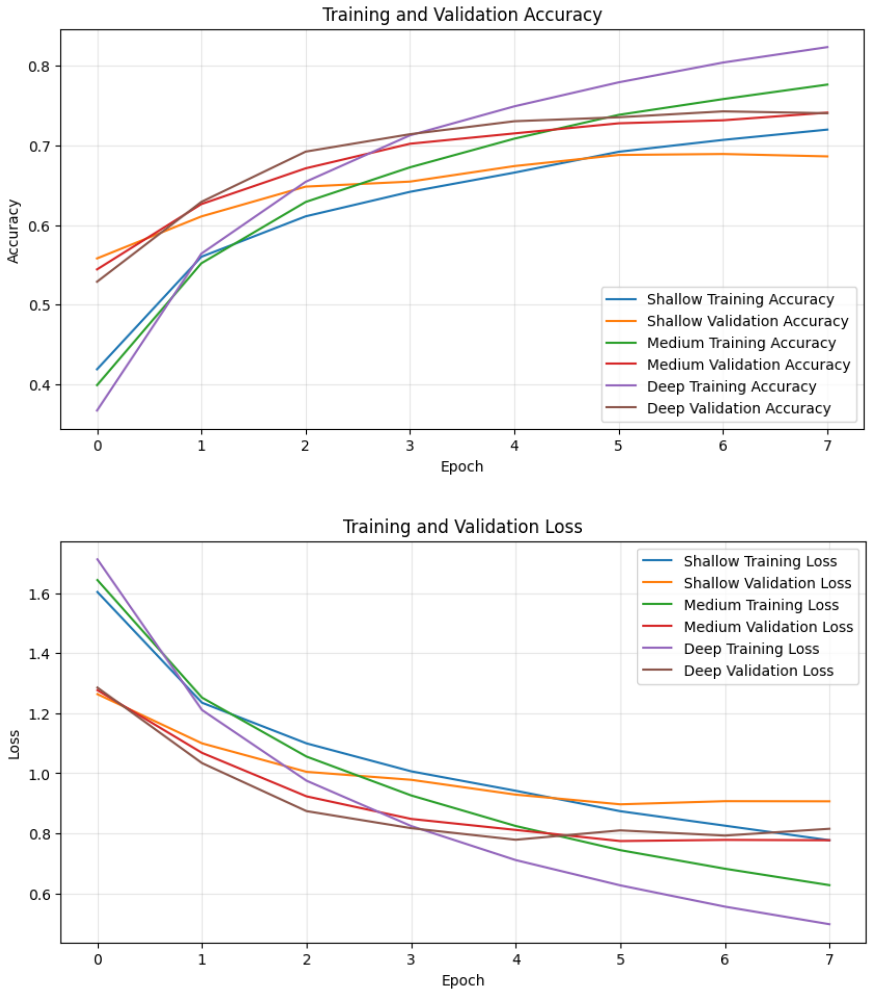
- Shallow CNN: A model with 2 convolutional layers designed to capture basic features such as edges and simple textures
- Medium CNN: A model with 4 convolutional layers designed to learn intermediate and structured feature representations
- Deep CNN: A model with 6 convolutional layers designed to learn more complex and abstract patterns

All models were trained using identical settings, including the Adam optimiser, categorical cross-entropy loss function, and eight training epochs. This ensures that differences in performance are primarily due to network depth rather than variations in training conditions.

RESULTS AND VISUALISATION

Training Performance

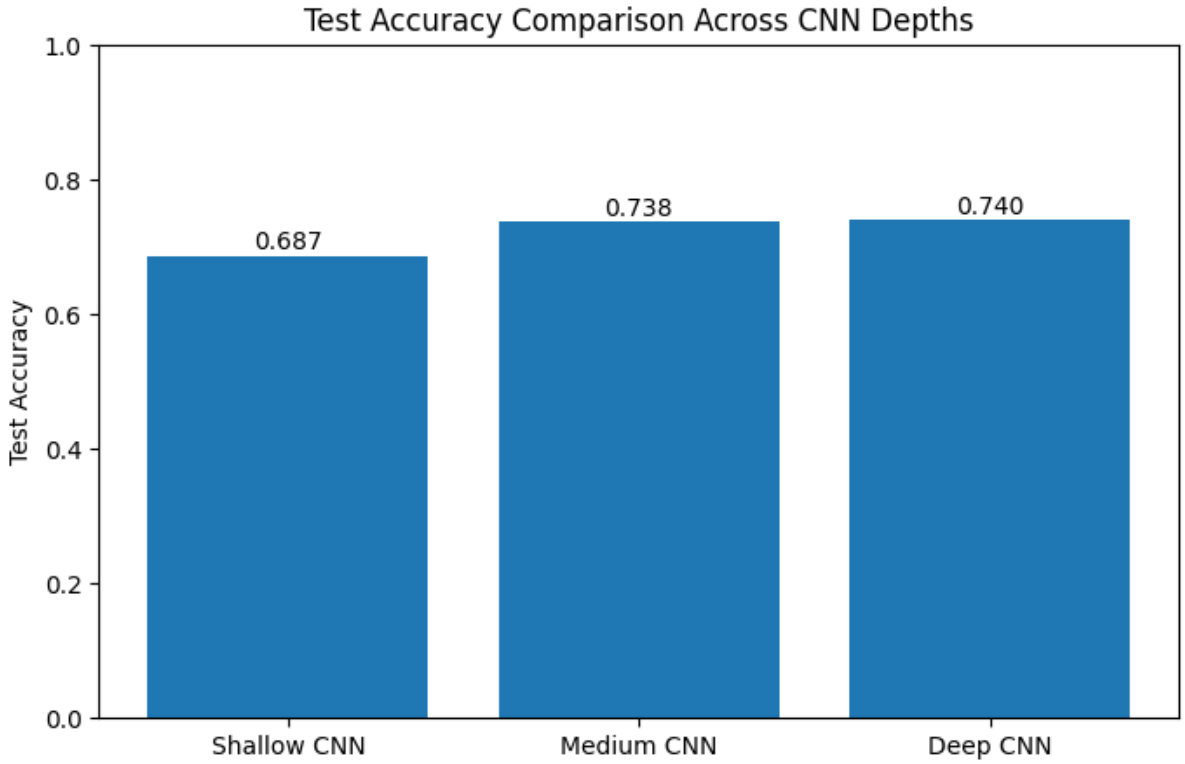
Figure 1: Training and validation accuracy and loss for shallow, medium, and deep CNN models across epochs.



Explanation:
The results show that performance improves as network depth increases. The shallow CNN achieves the lowest accuracy, while the medium CNN shows clear improvement, indicating better feature learning. The deep CNN achieves the highest accuracy, confirming its ability to learn more complex representations. However, the improvement from medium to deep is smaller, showing diminishing returns. A gap between training and validation accuracy suggests mild overfitting in deeper models, highlighting the trade-off between model complexity and generalisation.

Test Accuracy Comparison

Figure 2: Test accuracy comparison across shallow, medium, and deep CNN models.



Explanation:

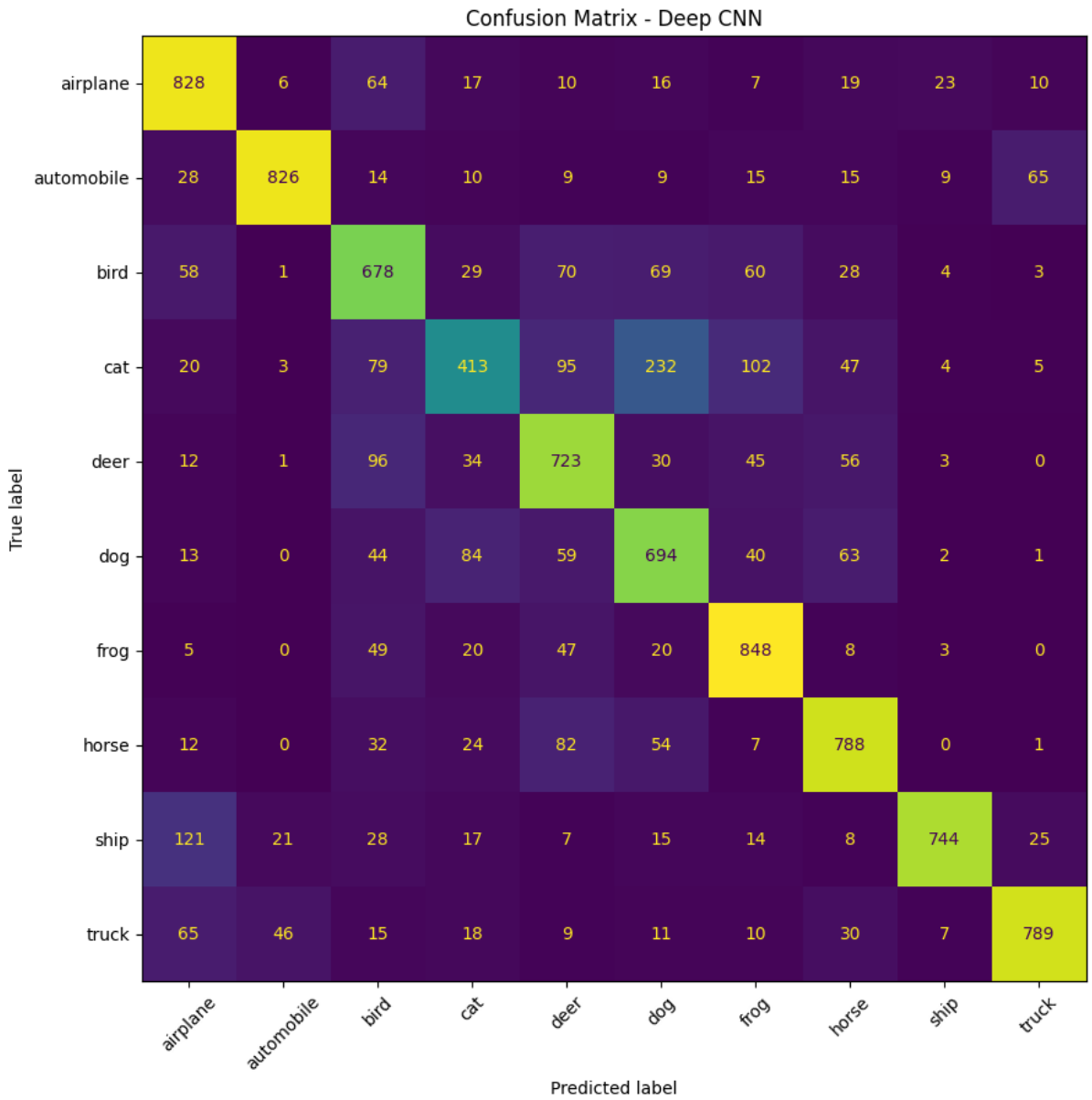
The bar chart provides a clear comparison of the final test accuracy achieved by the shallow, medium, and deep CNN models. The results show that accuracy increases as network depth increases, with the shallow CNN achieving the lowest performance, the medium CNN showing a significant improvement, and the deep CNN achieving the highest accuracy.

However, the improvement from the medium to deep CNN is relatively small compared to the improvement from shallow to medium. This indicates diminishing returns as network depth increases, where adding more layers provides smaller performance gains.

Overall, the chart highlights that while deeper models are more effective at learning complex feature representations, increasing depth beyond a certain point yields only marginal improvements in performance.

Confusion Matrix Analysis

Figure 3: Confusion matrix of the deep CNN model on the CIFAR-10 dataset.



Explanation:
The confusion matrix provides an overview of the model's performance across all classes. High values along the diagonal indicate strong classification accuracy for most categories. However, some misclassifications occur between visually similar classes, such as cat and dog or deer and bird. These errors highlight the limitations of feature learning, as similar patterns can be difficult to distinguish even for deeper networks. Although increased depth improves representation, some overlap between classes remains, affecting classification performance.

Classification Report

Figure 4: Classification report of the deep CNN model showing precision, recall, and F1-score for each class.

Classification Report - Deep CNN

Class	Precision	Recall	F1-Score	Support
Airplane	0.81	0.71	0.76	1000
Automobile	0.89	0.84	0.87	1000
Bird	0.70	0.58	0.63	1000
Cat	0.60	0.49	0.54	1000
Deer	0.65	0.76	0.70	1000
Dog	0.66	0.62	0.64	1000
Frog	0.84	0.78	0.81	1000
Horse	0.68	0.86	0.76	1000
Ship	0.80	0.87	0.83	1000
Truck	0.76	0.88	0.81	1000
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Accuracy			0.74	10000
Macro Avg	0.74	0.74	0.74	10000
Weighted Avg	0.74	0.74	0.74	10000

Explanation:
The classification report evaluates the model using precision, recall, and F1-score, with an overall accuracy of 73%, indicating good performance on the CIFAR-10 dataset. Some classes achieve strong results, while others show lower performance due to visual similarity between categories. For example, classes such as cat perform poorly compared to others, reflecting difficulty in distinguishing similar objects. The macro and weighted averages are both 0.73, indicating balanced performance across classes. Overall, the results show that while the deep CNN performs well, some categories remain challenging.

Feature Map Visualisation

Figure 5: Feature maps from shallow, medium, and deep CNN models.

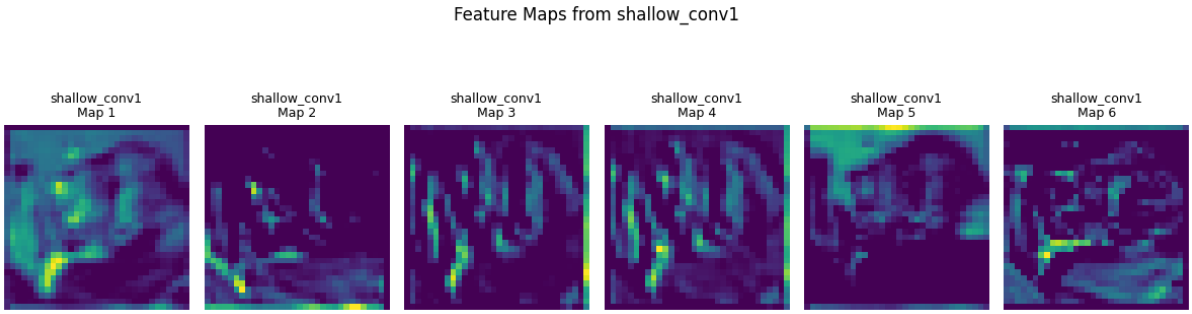
Original Image

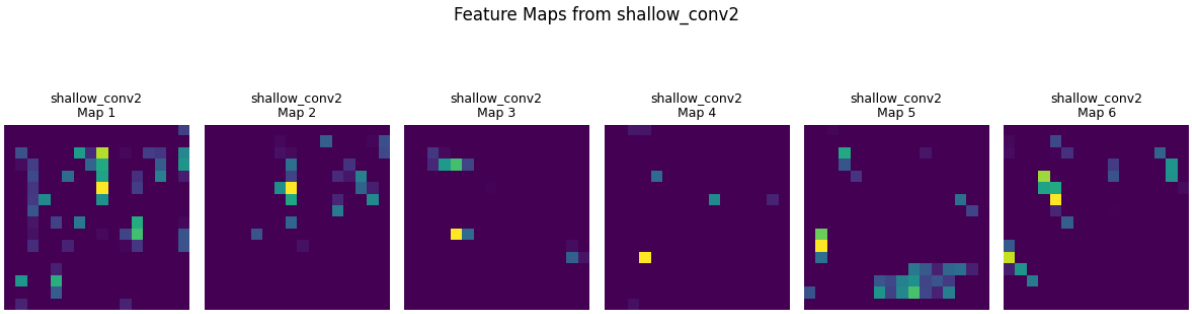
The input image from the CIFAR-10 dataset is used to illustrate how different CNN layers extract features.

Original Image: cat

Shallow CNN

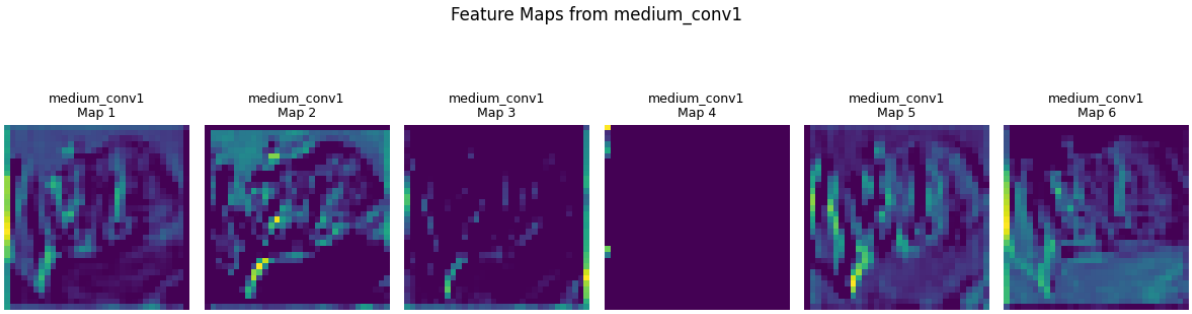
The shallow CNN captures basic patterns such as edges and simple textures. These features provide low-level information but lack the complexity needed to distinguish more challenging object categories.

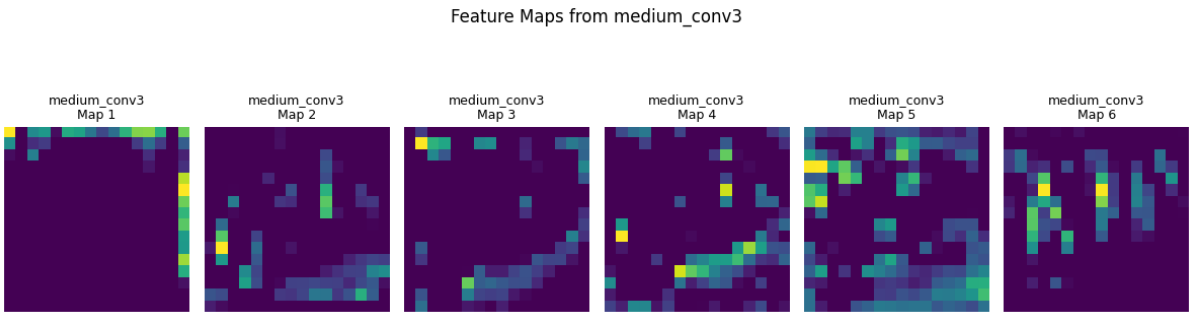


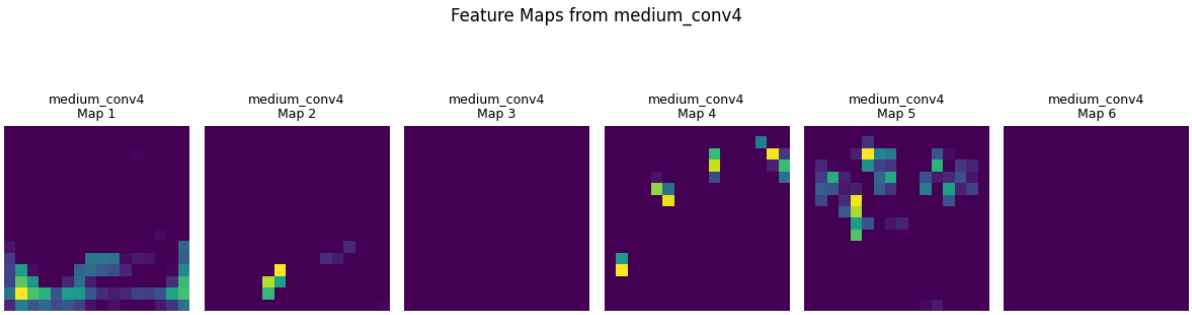


Medium CNN

The medium CNN represents an intermediate stage of feature learning. While early layers detect simple patterns, deeper layers combine them into more structured and meaningful representations. These features are more refined than those of the shallow model but less abstract than those in the deep CNN.

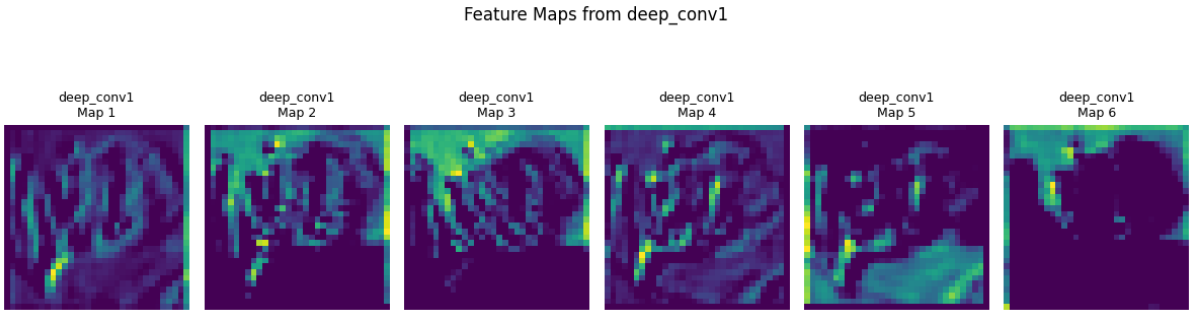


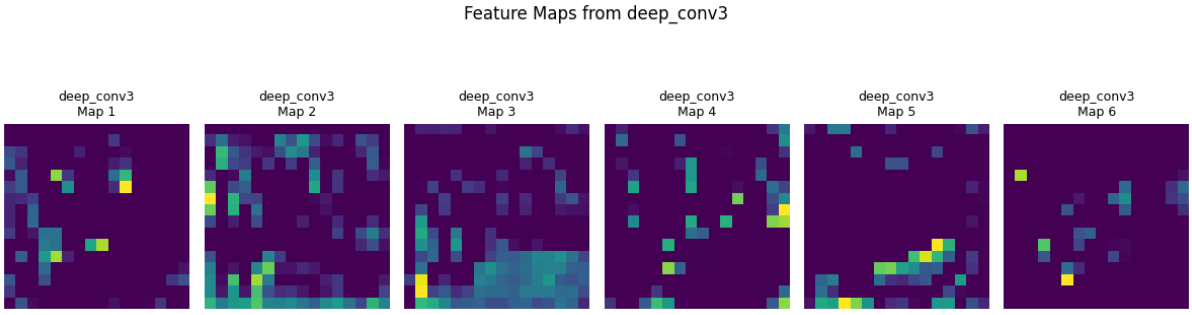


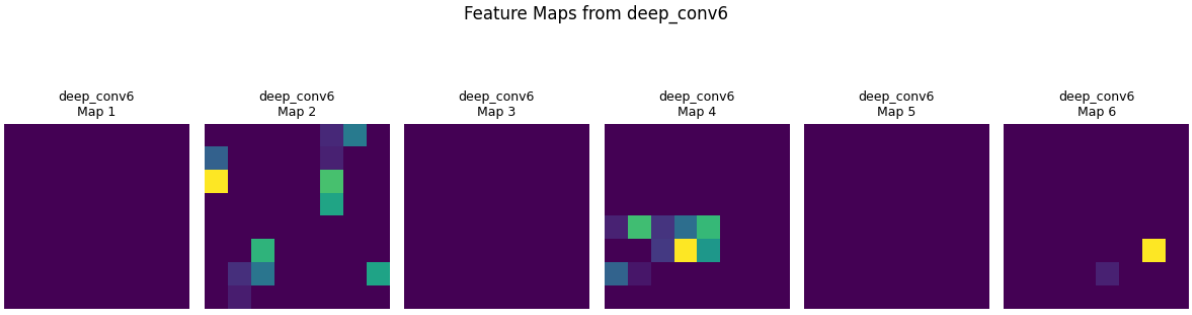


Deep CNN

The deep CNN shows a clear progression in feature learning. Early layers detect basic structures, while deeper layers produce highly abstract and selective representations that focus on the most relevant parts of the image.



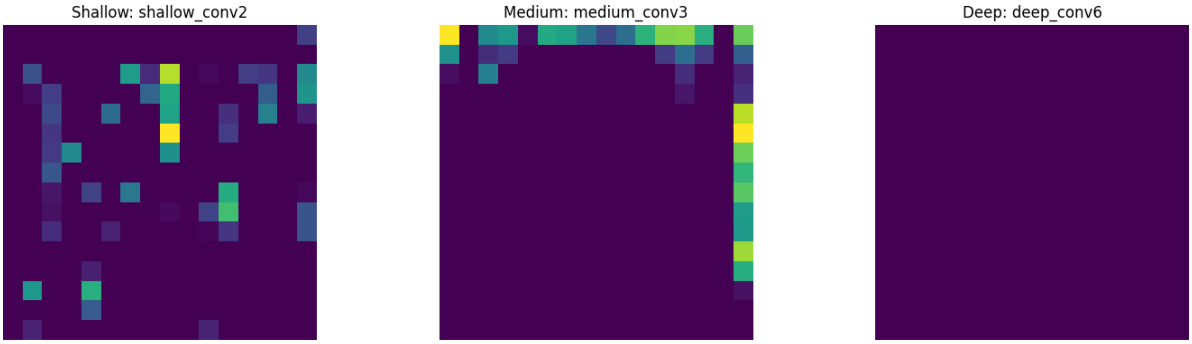




Explanation:
Feature maps demonstrate hierarchical learning in CNNs. Shallow layers capture simple features, while deeper layers produce more abstract representations. The medium CNN highlights the gradual transition between these stages.

Direct Feature Comparison

Figure 6: Direct comparison of feature maps across CNN depths.



A comparison of feature maps from the shallow, medium, and deep CNN models shows how representations evolve with depth. The shallow CNN produces scattered activations, capturing simple features. The medium CNN shows more structured patterns, indicating an intermediate stage of feature learning. In contrast, the deep CNN produces sparse and focused activations, highlighting only the most relevant regions. This demonstrates the hierarchical nature of CNNs, where increasing depth leads to more refined and abstract representations.

DISCUSSION

The results demonstrate that increasing network depth improves both feature learning and classification performance. The shallow CNN captures basic patterns, while the medium and deep CNNs learn progressively more complex and structured representations.

The inclusion of a medium-depth CNN shows that performance improves gradually with depth, rather than only when moving from shallow to very deep architectures. However, the improvement becomes smaller at higher depths, indicating diminishing returns.

Deeper models also introduce trade-offs. They require more computational resources, longer training times, and are more prone to overfitting, as observed in the training and validation performance gap.

Overall, these findings highlight the importance of selecting an appropriate network depth based on task complexity, balancing performance improvements with computational efficiency and generalisation.

PRACTICAL TAKEAWAYS

- Deeper CNNs generally achieve higher accuracy on complex datasets due to their ability to learn more abstract feature representations
- Medium-depth CNNs provide a balance between performance and computational cost, demonstrating that improvements occur progressively with depth
- Shallow models are faster and suitable for simpler tasks or resource-constrained environments
- Feature visualisation is a useful tool for understanding how models learn and interpret input data
- Model depth should be chosen based on task complexity, available computational resources, and the risk of overfitting

ETHICAL CONSIDERATIONS

Machine learning models can reflect biases present in the training data. The CIFAR-10 dataset may not fully represent real-world diversity, which can affect model fairness and lead to biased predictions.

Additionally, deeper models require more computational power, which has environmental implications due to increased energy consumption. As model complexity increases, so does the carbon footprint associated with training.

It is important to use these models responsibly, ensure fairness in predictions, and consider the environmental impact when designing and deploying machine learning systems.

ACCESSIBILITY CONSIDERATIONS

Several steps were taken to improve accessibility:

- Clear section headings were used to enhance readability and structure
- Figures were labelled and explained in accompanying text to support understanding
- Visualisations were designed to be interpretable without relying solely on colour, supporting colour-blind users
- A colour map (viridis) was used to improve contrast and readability
- Simple and clear language was used throughout to ensure the tutorial is accessible to a wider audience

CONCLUSION

This tutorial investigated how CNN depth affects feature learning and classification performance. The results demonstrate that increasing network depth improves both feature learning and accuracy, as deeper models capture more complex and abstract representations.

However, the benefits become more gradual at higher depths, and deeper models introduce challenges such as increased computational cost and risk of overfitting. Therefore, selecting an appropriate network depth requires balancing model complexity with task requirements and available resources.

Overall, CNN depth plays a crucial role in enabling hierarchical feature learning, making deeper networks more effective for complex image classification tasks.

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